AN ENHANCEMENT OF MULTI-OBJECTIVE WEIGHTED HUNGARIAN ALGORITHM FOR PRIORITY-BASED RELIEF GOODS DISTRIBUTION IN DISASTER-PRONE PHILIPPINE BARANGAYS

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Chapter ONE

INTRODUCTION

1.1 Background of the Study

The Philippines, situated within the Pacific "Ring of Fire" and the Typhoon Belt, experiences an average of 20 tropical cyclones annually, leading to frequent and severe displacement in local barangays (Pu et al., 2025). During the post-disaster response, the "last-mile distribution" of relief goods—matching specific aid packages to household needs—remains a critical logistical bottleneck. In the aftermath of recent typhoons like Odette and Kristine, local government units (LGUs) reported that despite having sufficient supplies, the lack of a structured allocation system resulted in "mismatched aid," where households with infants received standard food packs instead of necessary hygiene or medical kits (Ghahremani-Nahr et al., 2024).

The Hungarian Algorithm, also known as the Kuhn-Munkres algorithm, is a foundational combinatorial optimization method designed to solve the "Assignment Problem" by finding the minimum cost in a bipartite matching scenario (matching n resources to n recipients). While highly effective in static environments, the standard algorithm is restricted by a "single-objective cost matrix," typically focusing only on physical distance or travel time (Lim et al., 2024). Research by Luo et al. (2023) identifies a persistent pattern of inequity in these models, noting that they fail to account for household vulnerability factors—such as the presence of elderly members, persons with disabilities (PWDs), or infants—which should take priority over geographical proximity in a life-safety context.

The technical necessity for enhancing this algorithm is further underscored by the "stochastic nature" of disaster environments. Standard Hungarian implementations are mathematically "blind" to resource type compatibility, often treating all relief packs as identical units (Lim et al., 2024). Recent studies by Rodríguez-Espíndola et al. (2023) emphasize that a "Multi-Objective Weighting" approach—integrating Urgency Scores and Resource-Need Matching into the cost matrix—is essential to transform the algorithm from a simple distance-calculator into an intelligent decision-support tool. By enhancing the Hungarian Algorithm with these weighted variables, relief operations can move from a "first-come-first-served" basis to an optimized, priority-driven distribution system (Ghahremani-Nahr et al., 2024).

In addition to allocation challenges, disaster response systems are vulnerable to data integrity issues, particularly the presence of ghost beneficiaries or invalid records. Inaccurate vulnerability claims, such as false PWD declarations, can distort priority scoring and lead to inequitable resource allocation. Ensuring data authenticity is therefore essential for optimization-based systems.

Existing Algorithm: Standard Hungarian (Kuhn-Munkres) Baseline

A. Initialization and Matrix Construction

1. Identify the set of n relief resources (R) and n household recipients (H).
2. Construct an $n \times n$ cost matrix (C) where each element c_i, j represents the physical distance or travel time between resource i and household j. (Problem 1 and 2)

B. Matrix Reduction and Zero Identification

3. Subtract the minimum value of each row from every element in that row (Row Reduction).
4. Subtract the minimum value of each column from every element in that column (Column Reduction).
5. Identify all independent zero-cost assignments in the reduced matrix.

C. Iterative Covering and Pathfinding

6. Cover all zero elements in the matrix with the minimum possible number of horizontal and vertical lines (L). (Problem 3)

7. If L < n.

a. Find the smallest uncovered value (k).

b. Subtract k from all uncovered elements

c. Add k to elements covered by two lines.

d. Return to Step 6.

D. Final Assignment

8. Identify a set of n independent zeros where each row and column is assigned exactly once.

9. Output the final assignment schedule (S) for logistics dispatch.

1.2 Statement of the Problem

The primary problem of this study is the structural limitations of the existing Hungarian Algorithm when applied to complex allocation systems that require multiple optimization criteria and scalable computation. While the algorithm is widely used for solving classical assignment problems, its traditional formulation is designed primarily for single-objective optimization using a fixed cost matrix. As problem sizes increase and decision parameters become more complex, these structural characteristics limit the algorithm's flexibility and computational efficiency.

Specifically, this research seeks to address the following algorithmic problems:

1.2.1 The single-objective cost function of the Hungarian Algorithm limits its ability to consider multiple decision criteria during the assignment process.

The standard Hungarian Algorithm is designed for single-objective optimization, where each assignment is based on only one cost variable, typically distance or time. This limitation restricts its ability to handle complex decision environments where multiple criteria—such as household vulnerability, urgency, and resource compatibility—must be considered simultaneously. As a result, the algorithm cannot effectively represent multi-dimensional decision-making in disaster response scenarios.

Furthermore, the algorithm assumes that all input data are accurate and reliable. In real-world disaster contexts, however, beneficiary data may contain duplicates, inconsistencies, or unverifiable claims, leading to the inclusion of "ghost beneficiaries." These data integrity issues can distort priority weighting and compromise the accuracy and fairness of resource allocation.

Figure 1.1:

Comparison of assignment outcomes using the standard Hungarian algorithm with a single-objective cost function and a multi-criteria assignment approach.

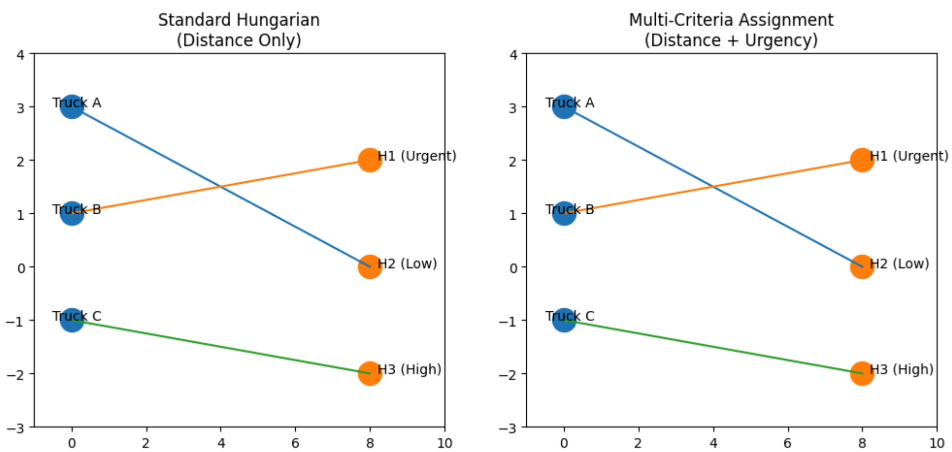


Figure 1.1. The left diagram illustrates the assignment produced by the standard Hungarian Algorithm when only distance is used as the cost parameter. Because the algorithm optimizes a single objective, resource allocation is determined solely by spatial proximity, without considering urgency levels. The right diagram illustrates a multi-criteria cost formulation that incorporates both distance and urgency. This comparison demonstrates that the single-objective structure of the Hungarian Algorithm limits its ability to represent multiple decision factors within the assignment process.

1.2.2 Requiring a fixed assignment matrix during execution reduces the algorithm’s ability to adjust to changing decision conditions.

The Hungarian Algorithm operates under a static cost matrix assumption, where all cost values must be predefined before the algorithm begins execution. Once the optimization process starts, the matrix values remain fixed throughout the computation, with no internal mechanism to detect or respond to changes in cost values once execution has commenced. This architectural constraint represents a fundamental flaw in the algorithm's design, as when cost changes do occur, the algorithm is forced to restart its entire initialization process from the beginning — rendering the solution ineffective in terms of both time and memory consumption. In systems where priorities or parameters may evolve during optimization, the static structure of the cost matrix prevents the algorithm from adapting to updated conditions, potentially compromising the accuracy and overall efficiency of the assignment solution.

Figure 1.2

Illustration of the static cost matrix limitation of the Hungarian algorithm in dynamic decision environments.

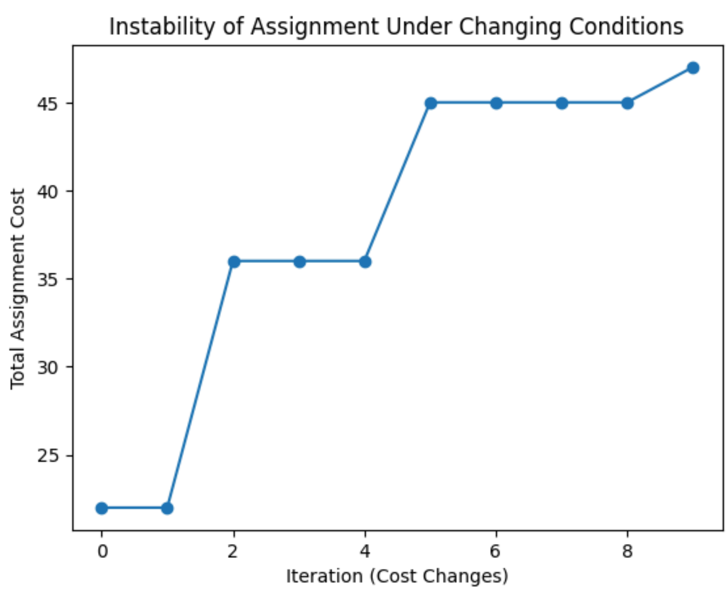


Figure 1.2. This graph shows how the total assignment cost changes as the cost matrix is updated over time. Each point represents a new computation after a cost change. Instead of maintaining a stable or gradually improving solution, the total cost fluctuates and sometimes increases significantly. This demonstrates that the Hungarian Algorithm cannot adapt to changes incrementally and must recompute from scratch, leading to inconsistent and inefficient results in dynamic environments.

1.2.3 The rapid growth of computational operations as matrix size increases leads to slower processing in large assignment problems.

The Hungarian Algorithm operates with a computational time complexity of $O(n^3)$, meaning that the number of required operations increases cubically with respect to the size of the assignment matrix. During the optimization process, the algorithm repeatedly performs matrix reductions, zero identification, and iterative adjustment procedures. While this complexity is manageable for smaller datasets, the computational workload increases significantly as the matrix dimension grows. As the number of agents and tasks expands, the repeated matrix operations can result in longer processing times, which may affect the algorithm's efficiency when solving large-scale assignment problems.

Figure 1.3

Algorithmic Processing Latency and Computational Complexity Across Varying Population Densities

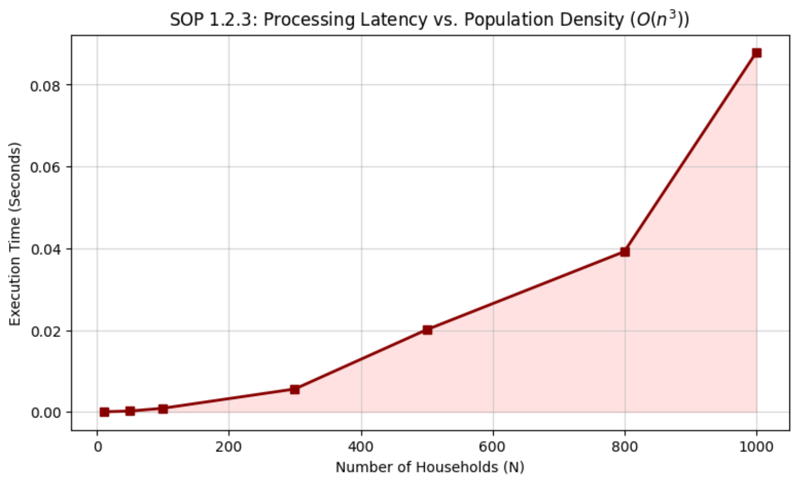


Figure 1.3 plots algorithm execution time against the number of households (N = 0 to 1,000) for the standard Hungarian Algorithm. The curve remains nearly flat at low population sizes but rises sharply beyond N = 400, reaching approximately 0.085 seconds at N = 1,000. This non-linear growth pattern directly confirms the $O(n^3)$ computational complexity flaw identified in SOP 1.2.3, where the algorithm provides no internal mechanism to limit or reduce the number of operations as the matrix scales — resulting in exponentially increasing processing demands that render the algorithm increasingly inefficient as the size of the assignment problem grows.

1.3 Objective of the Study

This study aims to develop a simulation-based disaster relief goods distribution system utilizing an Enhanced Multi-Objective Weighted Hungarian Algorithm to address the limitations of the standard Hungarian Algorithm in disaster-response operations within Philippine barangays. The proposed system integrates a three-variable cost matrix consisting of geographical distance, household urgency scores, and resource-type compatibility to support priority-driven, equitable, and computationally efficient relief allocation. Through the developed system, both the standard and enhanced Hungarian Algorithm are implemented and evaluated within a controlled simulation environment.

Specifically, the study aims to accomplish the following objectives:

1. To address the limitation of single-objective optimization by incorporating multiple decision criteria, such as geographical distance, household urgency, and resource-type compatibility, into the assignment cost formulation.
2. To design a weighted cost matrix structure that allows the integration of multiple priority factors within the assignment process.
3. To evaluate the computational performance of the enhanced algorithm in comparison with the standard Hungarian Algorithm when applied to larger assignment matrices using metrics such as Total Assignment Cost, Mean Allocation Accuracy, prioritization efficiency, and execution time.

1.4 Significance of the Study

The development of an Enhanced Multi-Objective Weighted Hungarian Algorithm carries direct and measurable implications for humanitarian logistics, disaster risk governance, and computer science research in the Philippine context. The following stakeholders stand to benefit most significantly from the findings and technical contributions of this study:

Barangay Officials and Local Government Units (LGUs). This study provides barangay officials and LGU disaster coordinators with a structured, data-driven allocation framework that automates the matching of categorized relief packs to profiled households. By eliminating manual sorting — a process highly susceptible to human error and administrative delay under crisis conditions — the algorithm enables LGU personnel to generate optimized distribution schedules rapidly, ensuring that limited resources reach the right households before recovery windows close.

Disaster Response Agencies (e.g., DSWD, NDRRMC, and the Philippine Red Cross). For regional and national disaster response agencies operating under resource constraints, the multi-objective weighting mechanism ensures that specialized relief packs — including medical kits, infant supplies, and mobility aids — are systematically routed to the most vulnerable recipients rather than the nearest ones. This directly addresses the documented pattern of aid mismatch in post-typhoon operations and increases the measurable utility of every distributed relief pack.

Vulnerable Residents and Households. The primary and most direct beneficiaries of this study are the residents of disaster-prone Philippine barangays — particularly the elderly, infants, and persons with disabilities (PWDs). The urgency scoring mechanism embedded in the enhanced algorithm ensures that these high-risk groups are structurally prioritized in every assignment cycle, effectively reducing their "Time-to-Care" and mitigating the compounding health risks associated with delayed or mismatched relief delivery.

Future Researchers and the Academic Community. This study contributes to the emerging field of humanitarian informatics by demonstrating a replicable methodology for enhancing classical combinatorial optimization algorithms with domain-specific weighting functions. The documented simulation framework, multi-objective cost matrix design, and performance benchmarking protocol serve as a reproducible foundation for future research exploring hybrid optimization models for disaster-specific assignment, routing, and resource allocation problems in high-density urban environments.

1.5 Scope and Limitations

This section defines the operational boundaries of the study and identifies the constraints that govern its implementation and generalization. The distinction between scope and limitations is deliberate: the scope describes what the study is designed and equipped to investigate, while the limitations acknowledge the conditions under which its findings must be interpreted.

The scope of this study is limited to the design, development, simulation, and performance evaluation of a disaster relief goods distribution system utilizing an Enhanced Multi-Objective Weighted Hungarian Algorithm for Philippine barangays. The system is designed to automate the allocation of categorized relief resources to verified household recipients using a multi-objective weighted cost matrix that integrates geographical distance, household urgency scores based on vulnerability factors (such as elderly members, persons with disabilities, and infants), and resource-type compatibility.

The developed system serves as a simulation and decision-support platform where both the standard Hungarian Algorithm and the enhanced weighted version are implemented and compared under controlled experimental conditions. The system allows the input of household and relief resource data, generation of assignment matrices, execution of allocation procedures, and presentation of comparative allocation results and performance metrics.

To ensure data integrity, a rule-based beneficiary validation mechanism is incorporated prior to optimization. Household records are cross-checked against a barangay masterlist, duplicate entries are identified using multi-field matching techniques, and vulnerability claims such as PWD status are verified through registry validation. Only verified records are included in the allocation process.

The system is implemented in Python using Google Colab with supporting scientific computing libraries such as SciPy, NumPy, and Pandas, and operates using open-source or synthetic datasets based on realistic disaster-response scenarios. System and algorithm performance are evaluated using quantitative metrics including Total Assignment Cost, Mean Allocation Accuracy, prioritization efficiency, and execution time.

The study is subject to several technical and operational limitations. First, the system is limited solely to relief-goods assignment and allocation optimization and does not include route optimization, delivery navigation, evacuation planning, inventory forecasting, or real-time disaster monitoring functionalities. Second, the simulation environment assumes that all household data, vulnerability profiles, and geographical coordinates are accurate and pre-verified prior to execution. Although the system incorporates a Dynamic Re-Assignment mechanism for urgency-score changes, it does not simulate real-time data acquisition issues such as communication failures, incomplete records, or live disaster disruptions.

Third, the re-optimization threshold used to trigger Dynamic Re-Assignment is manually defined at $\Delta \geq 2$ and remains fixed throughout system execution. The system does not implement machine learning or adaptive threshold adjustment mechanisms based on disaster severity or historical allocation patterns. Fourth, the developed system functions only as a simulation-based decision-support tool and is not intended for direct real-world deployment in live disaster-response operations. Finally, the beneficiary validation mechanism relies on available external registry records and rule-based verification procedures; therefore, the system may not completely eliminate highly sophisticated forms of fraudulent or falsified beneficiary information.

Despite these limitations, the study aims to demonstrate that integrating a multi-objective weighted allocation approach within a disaster relief distribution system can produce more efficient, equitable, and priority-sensitive assignment outcomes compared to the standard single-objective Hungarian Algorithm.

1.6 Definition of Terms

For clarity and consistency, the following terms are defined as used in this study:

Assignment Problem - A class of combinatorial optimization problems concerned with finding the optimal one-to-one pairing between two sets — in this study, between categorized relief packs and profiled household recipients — such that a defined cost function is minimized across all possible assignments.

Bipartite Graph - A mathematical graph structure consisting of two mutually exclusive sets of vertices — specifically, the set of relief resources and the set of household recipients — in which edges connect only vertices from different sets, representing feasible assignment pairings.

Combinatorial Optimization - A field of mathematical and computational research focused on identifying the most efficient solution from a finite set of discrete configurations — in this study, the lowest-cost, highest-utility assignment of categorized relief goods to recipient households.

Cost Matrix - A two-dimensional array in which each entry represents the composite weighted cost of assigning a specific relief resource to a specific household recipient, integrating geographical distance, household urgency score, and resource-type compatibility into a single numerical value.

Deprivation Cost - A formally defined humanitarian logistics metric quantifying the human welfare loss incurred when a household does not receive appropriate aid in a timely manner. In this study, deprivation cost is the primary measure of aid mismatch and is minimized through resource-type compatibility weighting**.**

Dynamic Re-Assignment — the mechanism that monitors urgency score changes after an initial assignment has been made and automatically triggers a re-execution of the algorithm when the change exceeds a defined threshold.

Euclidean Distance - A geometric metric used to compute the straight-line spatial distance between two coordinate points. In this study, it serves as the base distance variable within the multi-objective cost matrix, representing the physical proximity between a barangay distribution hub and a recipient household.

Hungarian Algorithm (Kuhn-Munkres Method) - A primal-dual combinatorial optimization algorithm that solves the Assignment Problem in polynomial time by iteratively reducing a cost matrix until a complete set of minimum-cost one-to-one assignments is identified. The standard implementation forms the computational baseline against which the enhanced model is evaluated.

Multi-Objective Optimization (MOO) - A mathematical framework for simultaneously optimizing two or more conflicting objective functions within a single model. In this study, MOO is applied to balance the competing objectives of minimizing delivery distance, maximizing urgency prioritization for vulnerable households, and ensuring resource-type compatibility.

Polynomial Time Complexity $O(n^3)$ - A formal computational classification indicating that the number of operations required by an algorithm grows as a cubic function of the input size n — where n represents the number of households or relief resources. This characteristic of the standard Hungarian Algorithm is the source of processing latency in high-density deployment scenarios.

Priority Weighting - A numerical coefficient applied within the cost matrix to amplify or reduce the relative assignment cost of a household based on its vulnerability profile. Higher priority weights are assigned to households containing elderly members, persons with disabilities, or infants, structurally ensuring their prioritization in the final assignment output.

Resource-Type Compatibility - A binary or scaled matching score that quantifies the alignment between the categorical contents of a relief pack (e.g., medical kit, infant pack, standard food pack) and the documented needs of a recipient household. A high compatibility score reduces the assignment cost; a low score increases it, discouraging mismatched assignments.

Re-Optimization Threshold — the minimum change in urgency score (Δ) required to trigger a new round of assignment. In this study, the default threshold is set at $\Delta \geq 2$.

Stochastic Variables - Environmental or situational factors characterized by randomness or unpredictability, such as fluctuating flood levels, sudden road blockages, or dynamic shifts in household population during active disaster events. In this study, stochastic variables are acknowledged as a limitation of the simulation environment, which operates under pre-verified, static input assumptions.

Urgency Score - A quantitative index assigned to each recipient household that reflects the level of immediacy with which relief goods must be delivered, based on assessed vulnerability factors including the age composition, disability status, and medical dependency of household members. Urgency scores are integrated into the cost matrix as dynamic priority weights.

Utility Loss - The measurable reduction in the effectiveness of a relief assignment that occurs when a household receives a resource category that does not correspond to its documented needs. In this study, utility loss is directly produced by the standard Hungarian Algorithm's single-objective distance-only optimization and is quantified as a reduction in Mean Allocation Accuracy.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a systematic review of related literature and studies that provide the foundations for the present research. The review is organized into two major components: related literature, which encompasses theoretical frameworks, conceptual models, and foundational discussions on the Hungarian Algorithm, multi-objective optimization, and humanitarian logistics; and related studies, which covers applied and empirical research that demonstrates the algorithm's use in various contexts. Each component is further divided into local and international sources to situate the study within both the Philippine and global research landscapes. The synthesis of these works establishes the academic basis for the proposed enhancement of the Hungarian Algorithm as a multi-objective decision-support tool for disaster resource allocation.

2.1 Related Literature

THE HUNGARIAN ALGORITHM: THEORETICAL FOUNDATIONS AND ALGORITHMIC PROPERTIES

The Hungarian Algorithm, also designated as the Kuhn-Munkres method, remains one of the most extensively examined combinatorial optimization procedures in operations research. Its foundational theory rests on the work of König and Egerváry, whose contributions in graph theory and bipartite matching were synthesized by Kuhn (1955) and refined by Munkres (1957). As summarized by Kahil et al. (2022) in their assessment presented at the First Australian International Conference on Industrial Engineering and Operations Management, the algorithm operates through a structured procedure of matrix row and column reductions, followed by iterative zero-covering and optimal assignment determination. Their work demonstrates that the classical Hungarian method reliably produces minimum-cost solutions for both balanced and unbalanced assignment problems, confirming the algorithm's polynomial time complexity of $O(n^3)$ across practical implementations.

The theoretical significance of the algorithm extends beyond mere computational efficiency. As described by the ScienceDirect overview (2024), the Hungarian algorithm is grounded in König's theorem, which establishes that in a binary matrix, the minimum number of rows and columns covering all ones equals the maximum number of independent ones. This duality relationship between covering and matching provides the mathematical guarantee of optimality that distinguishes the Hungarian method from heuristic approaches. The algorithm's reliance on primal-dual relationships, wherein row and column reductions correspond to dual variable updates, anticipates later developments in linear programming duality and network flow theory. This theoretical richness makes it not merely an applied tool but a foundational artifact of combinatorial mathematics.

Gabrovšek et al. (2020), publishing in the journal Mathematics, extended these theoretical foundations by investigating the k-assignment problem — a generalization in which a weighted k-partite graph replaces the standard bipartite structure. Their paper introduces five novel heuristics that generalize the Hungarian Algorithm to handle cases where $k \geq 3$ partitions must be simultaneously matched. These heuristics include both greedy and randomized local search variants that build upon the standard algorithm's merging and reduction procedures. Their numerical evaluations showed that the best-performing variants achieve solutions differing from the mathematical optimum by 0.1% or less, confirming practical near-optimality for large-scale instances. This generalization is theoretically significant for the present study because it demonstrates that the classical bipartite matching framework underlying the Hungarian Algorithm can be systematically extended to accommodate more complex relational structures, such as the multi-dimensional assignment problems that emerge when multiple vulnerability criteria must be balanced simultaneously.

Parida et al. (2023), publishing in a peer-reviewed Indian research journal, provided a computational treatment of the Hungarian Algorithm applied to manufacturing task scheduling. Their study demonstrates that the algorithm consistently yields globally optimal solutions across multiple test cases, and that these solutions are equivalent to results produced by enumeration and simplex methods — but with significantly fewer computational steps. This finding reinforces the algorithm's practical appeal in applied settings where computational resources are limited and time-sensitive decisions must be made. The authors also confirmed the algorithm's ability to handle unbalanced matrices through dummy variable augmentation, establishing a procedural baseline that is directly referenced in the methodology of the present study.

MODIFICATIONS, EXTENSIONS, AND IMPROVEMENTS OF THE HUNGARIAN ALGORITHM

The limitation of the standard Hungarian Algorithm to single-objective, deterministic cost matrices has motivated a broad literature on algorithmic modifications and extensions. These enhancements pursue several distinct goals: handling unbalanced assignment structures, incorporating fuzzy or uncertain cost data, extending to multi-objective formulations, and improving computational performance for large-scale problems. Taken together, they form the theoretical lineage within which the present study is situated.

Annapurna and Yesaswini (2023), publishing in the International Journal of Communication and Computer Technologies, proposed an improved version of the Hungarian Algorithm specifically designed for unbalanced assignment problems — cases in which the number of agents does not equal the number of tasks. Their modification eliminates the standard approach of inserting dummy rows or columns with zero costs, which they argue distorts the true cost landscape and can lead to suboptimal real-world decisions. Instead, they develop a modified reduction procedure that preserves the optimality condition without artificial augmentation. This is directly relevant to disaster resource allocation contexts, where the number of available supply points frequently differs from the number of demand areas, making the unbalanced case the norm rather than the exception.

Similarly, a study published in the Global Journal of Science Frontier Research (2022) presented an improved Hungarian Algorithm for a specific class of unbalanced assignment problems characterized by asymmetric cost structures. The authors demonstrated that under these conditions, the traditional dummy-variable augmentation produces mathematically correct but operationally inefficient assignments. Their improvement focuses on modifying the zero-covering phase to account for asymmetry in the cost matrix, producing assignments that better reflect real-world constraints. The work is significant because it illustrates how even incremental modifications to the algorithm's internal logic can substantially improve solution quality in applied contexts.

Elsisy, El Saadany, and El Sayed (2020), publishing in the journal Complexity, addressed the challenge of fuzzy cost parameters — a condition that is endemic in disaster logistics, where distance estimates, travel times, and delivery costs are inherently imprecise. Their algorithm transforms fuzzy assignment problems into interval assignment problems using alpha-cut operations and interval arithmetic, after which a modified Hungarian method is applied to the resulting crisp intervals. This approach preserves the intrinsic uncertainty in cost data rather than forcing premature defuzzification, producing solutions that are more robust to parameter variation. The authors demonstrated the method's applicability through a real-world rehabilitation planning problem in Egypt. This work is foundational for the present study's argument that the Hungarian Algorithm must be extended to handle uncertainty if it is to be effective in disaster response settings.

Nandan, Tripathi, and Tiwari (2024), publishing in the Engineering and Technology Journal, advanced this line of research by proposing an optimized fuzzy Hungarian method that integrates robust ranking techniques for triangular fuzzy numbers. Their approach converts fuzzy cost coefficients into crisp equivalents using a ranking function that preserves the relative ordering of fuzzy values, enabling the standard Hungarian reduction procedure to be applied without loss of fuzzy information.

Comparative analysis with other fuzzy assignment methods demonstrated that their approach produces superior solutions in terms of total cost minimization while requiring fewer computational iterations. This efficiency gain is particularly relevant in time-critical humanitarian operations.

Wang et al. (2023), publishing in Geo-Spatial Information Science, proposed an improved Hungarian Algorithm-based task scheduling strategy for distributed computing environments handling large volumes of remote sensing data. Their modification introduces a load-balancing mechanism that supplements the cost matrix with resource utilization weights, effectively transforming the single-objective framework into a weighted two-objective model. The resulting algorithm produces assignments that minimize task completion time while maintaining equitable distribution of computational load across processing nodes. Although the application domain differs from humanitarian logistics, the architectural principle — augmenting the Hungarian cost matrix with secondary weight factors to achieve multi-criteria balance — is directly transferable to the present study's design of a vulnerability-weighted cost matrix.

Kozachko et al. (2023), publishing in the Expert Systems with Applications journal, investigated a hybrid approach that integrates the Hungarian Algorithm with genetic optimization for employee task assignment in vehicle service stations. Their method first generates a near-optimal solution using the Hungarian Algorithm and then applies genetic operators — crossover and mutation — to explore neighboring solutions and escape local optima. The proposed evaluation criterion incorporates worker qualifications, task complexity, and estimated completion time as composite dimensions of a single weighted score, enabling the method to handle heterogeneous agents with varying capabilities. This hybridization strategy is noteworthy because it demonstrates how the Hungarian Algorithm can serve as a backbone for more complex multi-criteria systems without sacrificing its core optimality guarantees.

MULTI-OBJECTIVE OPTIMIZATION IN ASSIGNMENT PROBLEMS

The single-objective structure of the classical Hungarian Algorithm has been widely recognized as insufficient for real-world assignment problems involving competing and incommensurable criteria. The multi-objective assignment problem (MOAP) literature has accordingly developed a rich body of methods for extending assignment optimization to accommodate multiple simultaneous objectives.

Chakraborty, Jana, and Roy (2022) proposed a direct extension of the Hungarian Algorithm to the multi-objective assignment problem, published in the Journal of Xi'an Shiyou University. Their methodology introduces an improved Hungarian Algorithm that applies the standard reduction procedure iteratively across multiple objective functions, using a weighted goal programming framework to convert the multi-objective problem into a sequence of single-objective sub-problems. The weights assigned to each objective reflect decision-maker priorities, and the algorithm searches for the compromise solution that minimizes the aggregate weighted deviation from the individual optima. The authors validated their approach on transportation and assignment problems with competing time, cost, and quality objectives. This work is the closest international algorithmic precedent for the present study, which similarly proposes to transform the single-objective Hungarian cost matrix into a weighted, multi-dimensional structure incorporating efficiency and vulnerability dimensions.

Tran and Nguyen (2023), publishing in the Journal of Economic Statistics, applied an improved multi-objective Hungarian Algorithm to the labor allocation problem in a manufacturing firm with fewer employees than available jobs. Their approach evaluates each potential worker-job assignment across multiple performance dimensions — productivity, skill compatibility, and workload balance — aggregated into a composite score through a multi-criteria decision function. The improved algorithm generates optimal assignments that outperform single-objective solutions on at least two of the three criteria in all tested scenarios. The study is particularly relevant because it demonstrates that multi-objective extensions of the Hungarian Algorithm are not only theoretically sound but also operationally superior to classical approaches in resource-constrained environments.

The theoretical framework for multi-objective stochastic optimization more broadly was synthesized in a comprehensive systematic review by Ghasemi and colleagues (2022), published in Applied Intelligence. Their review of over 570 peer-reviewed papers from 2018 to 2025 identifies the assignment problem as one of the most frequently studied domains in multi-objective stochastic optimization, particularly in humanitarian logistics and healthcare supply chain management. The authors highlight that stochastic multi-objective models are consistently superior to deterministic single-objective approaches in environments characterized by demand uncertainty, infrastructure disruption, and time pressure — precisely the conditions that define disaster response operations. This review provides broad empirical support for the theoretical direction of the present study.

HUMANITARIAN LOGISTICS: THEORY, EQUITY, AND OPTIMIZATION

Humanitarian logistics has emerged as a distinct domain within operations research, characterized by the integration of efficiency and equity objectives under conditions of extreme uncertainty. The theoretical literature in this area provides the conceptual scaffolding for the present study's argument that the standard Hungarian Algorithm, while mathematically optimal, is operationally inadequate for disaster response without substantial modification.

Rodríguez-Espíndola, Dey, Albores, and Chowdhury (2023), publishing in the Annals of Operations Research, developed a two-stage stochastic multi-objective programming model for humanitarian logistics that explicitly incorporates intermodal transportation and carbon emission constraints alongside cost minimization. Their model demonstrates that when decision-makers optimize only for cost efficiency, they systematically underserve peripheral communities with limited road access — a finding that parallels the equity critique of the standard Hungarian Algorithm. Their solution approach uses an epsilon-constraint method to generate Pareto-optimal frontiers, enabling decision-makers to select the most contextually appropriate trade-off between competing objectives. The concept of Pareto-optimality is directly applicable to the present study's approach of generating assignment solutions that balance logistical efficiency against household vulnerability.

Temiz et al. (2025), publishing in the International Transactions in Operational Research, proposed a probabilistic bi-objective model for humanitarian location-routing under uncertain demand and road closures. Their model addresses both the pre-positioning of relief supplies and the dynamic routing of delivery vehicles under stochastic conditions. The dual objectives — minimizing total logistics cost and minimizing maximum unmet demand — are solved using an epsilon-constraint formulation that generates multiple Pareto-optimal solutions. The authors' treatment of road closure uncertainty as a probabilistic event rather than a deterministic parameter is particularly relevant to Philippine disaster contexts, where typhoon-induced flooding routinely renders standard road networks inaccessible during precisely the period when relief distribution is most urgent.

Li, Zhang, and Yao (2024), publishing in the Journal of Cleaner Production, developed a bi-level robust optimization model for the coupled allocation of post-disaster personnel and material assistance. Their model separates strategic pre-disaster allocation decisions from tactical post-disaster distribution decisions, using robust optimization to hedge against worst-case demand realizations. The inclusion of a deprivation cost function — a penalty term that quantifies the welfare loss suffered by unserved beneficiaries — is theoretically significant because it introduces a social equity dimension into the optimization objective without requiring subjective weighting of incommensurable criteria. This deprivation cost concept provides a formal mechanism for incorporating household vulnerability into the assignment cost matrix, which is a central design feature of the present study.

Boostani, Jolai, and Bozorgi-Amiri (2021), publishing in the International Journal of Sustainable Transportation, designed a sustainable humanitarian relief logistics model for pre- and post-disaster phases that integrates social, economic, and environmental objectives. Their three-objective model minimizes total cost, maximizes coverage of demand areas, and minimizes carbon emissions from transportation. The inclusion of social and environmental dimensions alongside traditional cost metrics represents a fundamental reconceptualization of what it means for a humanitarian logistics system to be optimal. The authors note that single-objective cost minimization systematically disadvantages vulnerable populations in remote areas, who are both harder to reach and more dependent on external relief. This observation directly motivates the present study's incorporation of a vulnerability index into the assignment cost structure.

Cao, Yang Liu, and Tang (2021), publishing in the International Journal of Production Economics, proposed a fuzzy bi-level optimization model for multi-period post-disaster relief distribution in sustainable humanitarian supply chains. Their model handles demand uncertainty through triangular fuzzy membership functions and generates dynamic allocation plans across multiple time periods following a disaster event. The bi-level structure separates the decisions of a central relief coordinator from those of local distribution managers, capturing the hierarchical nature of real-world disaster response organizations. This work is relevant to the present study because it demonstrates that the assignment of resources to demand points must be reconsidered at each time step as conditions evolve — a dynamic dimension that the standard Hungarian Algorithm, operating on a static cost matrix, is inherently unable to address.

A research article published in PubMed Central (PMC) investigates the disproportionate impact of disasters on vulnerable populations, particularly persons with disabilities. The study documents that individuals with disabilities experience higher mortality rates, increased exposure to risk, and greater barriers to evacuation and access to emergency services. It also identifies structural and environmental challenges, including mobility limitations and communication barriers, which significantly hinder their ability to respond effectively during disaster events. These findings provide strong empirical support for prioritizing such groups within disaster-response systems. In the context of the present study, this evidence justifies the assignment of higher urgency weights to households with persons with disabilities, as it reflects their increased need for assistance and reduced capacity for self-evacuation or resource access.

The Sphere Handbook: Humanitarian Charter and Minimum Standards provides globally recognized guidelines for humanitarian response, emphasizing the principle of prioritizing assistance based on need and vulnerability. The framework identifies specific groups—including older persons, children, and persons with disabilities—as being at heightened risk during crises and requiring targeted support. It further stresses that humanitarian assistance must not only be efficient but also equitable, ensuring that the most vulnerable populations receive appropriate and timely aid. These standards reinforce the importance of incorporating vulnerability considerations into allocation mechanisms. The present study operationalizes this principle by integrating urgency scores into the cost matrix, thereby aligning algorithmic decision-making with established humanitarian practices.

According to the World Health Organization (WHO) Disability and Health Fact Sheet, persons with disabilities represent a population group with significantly higher health needs and reduced access to essential services, particularly during emergencies. The report highlights that individuals with disabilities are more likely to encounter barriers in mobility, communication, and access to healthcare, which are further exacerbated in disaster contexts. As a result, they face increased risks and require additional support compared to the general population. These findings underscore the necessity of prioritizing disability-inclusive approaches in disaster-response systems. In this study, such principles are reflected in the urgency scoring mechanism, where disability-related factors are assigned higher weights to ensure that allocation decisions account for elevated vulnerability levels.

A study accessed through EBSCOhost Academic Database examines disaster management frameworks with a focus on vulnerability-based planning and resource allocation strategies. The study highlights that effective disaster response systems must incorporate population heterogeneity, particularly accounting for groups with elevated risk profiles such as persons with disabilities, elderly individuals, and dependent populations. It further emphasizes that traditional allocation models relying solely on logistical efficiency are insufficient, as they fail to address disparities in need severity across affected populations. The findings support the integration of vulnerability-sensitive parameters into allocation models, reinforcing the need for multi-objective optimization approaches that balance efficiency with equity. This aligns with the present study's use of urgency-based weighting in the cost matrix to reflect differential vulnerability among households.

EMERGENCY RESPONSE OPTIMIZATION: RECENT ADVANCES

A comprehensive survey of recent advances in disaster emergency response planning, published in Science of Emergency Scenarios (2025), reviewed over two decades of research across five core areas: evacuation planning, facility location, casualty transport, search and rescue, and relief distribution. The survey, covering research from 2019 to 2024, identifies a consistent trend toward integrated optimization approaches that combine mathematical programming with machine learning and simulation. Among the findings most relevant to the present study, the authors note that assignment-based models — including those derived from the Hungarian Algorithm — continue to serve as the backbone of relief distribution optimization, but that their single-objective formulations are increasingly being supplemented with equity, urgency, and robustness considerations. The survey also confirms that reinforcement learning is emerging as a promising complement to classical assignment algorithms for dynamic, real-time allocation problems.

A two-stage robust optimization model for emergency service facility location-allocation published in Scientific Reports (2025) addressed demand uncertainty through a generalized budget uncertainty set that characterizes worst-case demand distributions. The model's three-dimensional objective function integrates facility setup cost, deprivation cost from the beneficiary perspective, and environmental impact cost in the response phase. The authors validated their model against data from simulated disaster scenarios and demonstrated that their robust solution outperforms deterministic equivalents in terms of worst-case service level. The explicit inclusion of deprivation cost as a distinct objective function component — rather than as a side constraint — provides a formal template for quantifying the social cost of inequitable allocation, which the present study adapts in its design of a vulnerability-weighted assignment matrix.

2.1.2 Local Related Literature

DISASTER RISK REDUCTION POLICY AND RESOURCE ALLOCATION IN THE PHILIPPINES

The Philippines occupies one of the most hazardous geographic positions in the world, situated within the Pacific Ring of Fire and directly in the path of the Western Pacific typhoon belt. This chronic exposure to natural hazards creates an ongoing and urgent demand for efficient, equitable mechanisms for disaster resource allocation. The institutional and policy landscape that governs this allocation provides the contextual framework within which the present study's algorithmic contribution must be understood.

The National Disaster Risk Reduction and Management Plan (NDRRMP) 2020–2030, published by the National Disaster Risk Reduction and Management Council (NDRRMC, 2020), establishes the overarching framework for disaster response in the Philippines. The plan adopts a wghole-of-society approach that recognizes local government units (LGUs), civil society organizations, the private sector, and academic and research institutions as co-producers of disaster resilience. For the present study, the NDRRMP is significant in two respects. First, it formally designates resource allocation as a core function of disaster response management, establishing the operational context for any algorithmic optimization model. Second, it explicitly acknowledges that vulnerable sectors — including persons with disabilities, the elderly, and households with medical dependencies — must be prioritized in the distribution of relief assistance. This vulnerability-prioritization mandate provides direct policy justification for the present study's incorporation of a vulnerability index into the Hungarian Algorithm's cost matrix.

The World Bank's report on the economic impact of disasters in the Philippines (World Bank, 2020) quantifies the scale of the resource allocation challenge. The report estimates that the Philippines loses an average of 0.5% of GDP annually to natural disasters, with significantly higher proportional losses in poorer provinces where households have fewer resources to absorb shocks independently. The report further notes that the inequitable distribution of government relief assistance — attributable in part to the absence of systematic, data-driven allocation tools — disproportionately affects the most vulnerable communities. This finding provides empirical grounding for the present study's argument that optimization models must explicitly incorporate equity considerations rather than defaulting to pure cost minimization.

The Department of Interior and Local Government's (DILG, 2021) Disaster Response Framework for Local Governments articulates the operational procedures through which LGUs are expected to mobilize and distribute relief resources in the immediate aftermath of a disaster. The framework identifies the assignment of relief goods to distribution points as a critical decision node in the response timeline, one that is currently executed through informal, experience-based processes rather than systematic optimization. This observation directly motivates the present study: by formalizing the assignment decision through an enhanced Hungarian Algorithm, the study aims to provide LGUs with a computationally tractable and contextually appropriate decision-support tool.

The Department of Science and Technology (DOST, 2020) has promoted the adoption of AI-driven technologies for disaster management as part of its broader science and technology agenda for disaster resilience. This policy positioning establishes a favorable institutional environment for algorithmic approaches to disaster resource allocation, signaling that the Philippine government regards computational optimization as a legitimate and desirable component of the national disaster response apparatus. The present study aligns with this institutional direction by proposing a specific, implementable algorithmic enhancement that can be operationalized within existing LGU data and administrative capacities.

Santos (2021), writing in the Journal of Emergency Services, examined the role of inter-agency coordination in determining the effectiveness of disaster response in the Philippines. His analysis identifies the absence of standardized, shared data systems as a primary barrier to efficient resource allocation, noting that relief goods are frequently duplicated in some areas while being entirely absent from others. The algorithmic approach proposed in the present study — which requires a systematically constructed cost matrix incorporating both logistical and vulnerability data — implicitly addresses this coordination problem by creating a common computational framework that integrates inputs from multiple agencies.

Ramos (2021), in a policy brief published by the Asian Development Bank Institute, documented the macroeconomic consequences of natural disasters in the Philippines and assessed the adequacy of existing policy responses. Among his findings, Ramos notes that post-disaster relief distribution tends to prioritize accessibility over need — meaning that communities located near roads and distribution centers receive more relief goods, regardless of their relative vulnerability, while geographically isolated communities receive less. This systematic access bias is precisely the kind of inequitable allocation outcome that the present study's vulnerability-weighted Hungarian Algorithm is designed to correct.

Baltazar, Florencio, Vicente, and Belizario (2024), publishing in the International Journal of Artificial Intelligence, examined the role of AI in enhancing disaster prediction, risk management, and mitigation in the Philippines through qualitative interviews with disaster management officials, meteorologists, and researchers conducted between June 2023 and March 2024. Their findings highlight that AI technologies, including machine learning and neural network models, have significantly improved disaster forecasting accuracy in the Philippines. More directly relevant to the present study, the authors note a gap between the maturity of disaster prediction capabilities and the relative underdevelopment of AI-supported resource allocation and distribution tools. This gap confirms the theoretical and practical significance of applying algorithmic optimization to the post-disaster resource assignment problem in the Philippine context.

Villanueva (2022) examined the current state and future potential of AI in disaster response, identifying the optimization of relief distribution as one of the most impactful areas for algorithmic intervention. His work emphasizes that the computational complexity of real-world disaster allocation — involving hundreds of supply points, thousands of demand nodes, and multiple competing objectives — exceeds the capacity of human judgment operating without formal decision-support tools. This observation provides intellectual justification for the present study's algorithmic approach and positions the enhanced Hungarian Algorithm as a practically deployable component of a broader AI-supported disaster response system.

2.2 Related Studies

ALGORITHMIC STUDIES ON THE HUNGARIAN METHOD AND ASSIGNMENT PROBLEMS

The empirical literature on the Hungarian Algorithm and its modifications spans several distinct application domains, each of which illuminates a different dimension of the algorithm's capabilities and limitations. The present review focuses on studies that directly extend, modify, or empirically evaluate the algorithm's performance, as these are most directly relevant to the methodological contributions of the present research.

Ning, Li, Yao, and Zhang (2024), publishing in Computers and Industrial Engineering, proposed the Advanced Hungarian Algorithm (AHA) for worker assignment in assembly production lines. Their modification introduces a tie-pool mechanism that stores and compares competing assignments when multiple workers are equally qualified for a given task, enabling the algorithm to break ties systematically rather than arbitrarily. The AHA was benchmarked against standard branch-and-bound and heuristic methods on a set of industrial test cases and demonstrated superior computational efficiency while maintaining optimality guarantees. The tie-pool concept is theoretically relevant to disaster allocation contexts, where multiple supply points may be equidistant from a demand node and secondary criteria — such as stock capacity or vehicle availability — must serve as tiebreakers.

Gabrovšek et al. (2020), whose theoretical contributions were discussed in the Related Literature section, also conducted extensive computational experiments comparing five heuristic variants of the Multiple Hungarian Method on both uniform and non-uniform weight distributions. Their results showed that the more sophisticated variants — designated C, E, and F in their taxonomy — substantially outperform simpler greedy variants on non-uniformly distributed instances, and that performance differences grow with instance size. This empirical finding is significant because real-world disaster allocation problems are characterized by highly non-uniform demand distributions, suggesting that more sophisticated algorithmic variants will be necessary for effective performance.

Chakraborty, Jana, and Roy's (2022) empirical evaluation of their improved Hungarian Algorithm for multi-objective assignment problems included comparative testing against weighted sum, goal programming, and Pareto-optimal enumeration methods. Their results demonstrated that the improved Hungarian approach achieves solution quality comparable to full Pareto enumeration at a fraction of the computational cost, making it practical for real-time decision-making. The study also showed that the weighted goal programming framework is sensitive to the choice of weights, confirming that domain-specific knowledge — such as vulnerability assessment data — is essential for correctly parameterizing the multi-objective cost function. This empirical finding underscores the importance of the vulnerability index design component in the present study.

Tran and Nguyen (2023) empirically validated their multi-objective improved Hungarian Algorithm on labor allocation data from a Vietnamese manufacturing firm with 12 workers and 15 jobs. Their results showed that the multi-objective solution consistently outperformed the single-objective Hungarian solution on at least two of the three evaluated dimensions — productivity, skill compatibility, and workload balance — without incurring any degradation on the remaining dimension. This Pareto-dominance result is a strong empirical demonstration that multi-objective extension of the Hungarian Algorithm produces genuinely superior solutions in practice, not merely in theory.

Kozachko et al. (2023) validated their hybrid genetic-Hungarian approach on task assignment data from vehicle service stations in Ukraine, comparing the hybrid method against three baseline algorithms: the standard Hungarian Algorithm, a standalone genetic algorithm, and a simulated annealing heuristic. The hybrid approach produced the best average solution quality across all test instances, with particularly strong advantages on larger instances where the genetic operators were able to explore a wider solution neighborhood than the deterministic Hungarian phase alone. The practical implication is that the Hungarian Algorithm, while globally optimal for single-objective problems, may produce locally suboptimal solutions when extended to multi-objective settings — a finding that motivates the careful design of weighting and aggregation mechanisms in the present study.

Wang et al. (2023) validated their improved Hungarian Algorithm for remote sensing task scheduling on real distributed computing clusters, demonstrating that the load-balancing modification reduced average task completion time by 18.3% compared to the standard Hungarian approach while simultaneously improving resource utilization rates. The experiment used matrices of up to 500 by 500 dimensions, confirming the algorithm's scalability to problem sizes substantially larger than those typically encountered in LGU-scale disaster allocation. This scalability result is encouraging for the present study, as it suggests that the enhanced algorithm will remain computationally tractable even as the size of the cost matrix grows with the number of supply points and demand areas.

Elsisy, El Saadany, and El Sayed (2020) validated their interval-Hungarian approach on a building rehabilitation case study in Egypt involving 20 contractors and 20 construction tasks with fuzzy cost estimates. Their algorithm produced optimal interval assignments in all 20 test configurations, and the resulting assignment costs were verified to lie within the expected fuzzy cost intervals in every case. Importantly, the approach required no defuzzification — preserving the full uncertainty structure of the original cost data — and achieved computational efficiency comparable to the standard crisp Hungarian method. This empirical result is directly relevant to the present study: in disaster scenarios where household vulnerability assessments involve inherent measurement uncertainty, an interval or fuzzy extension of the assignment cost matrix will preserve more information than a point estimate.

EMPIRICAL STUDIES IN HUMANITARIAN LOGISTICS OPTIMIZATION

Beyond the specifically algorithmic literature, a body of empirical research has tested various optimization approaches to humanitarian logistics, providing benchmarks against which the present study's contributions can be evaluated.

Ghasemi, Goodarzian, and Abraham (2022), publishing in Applied Intelligence, developed and tested a stochastic multi-objective location-allocation-routing model for earthquake relief logistics using real data from earthquake-affected regions. Their model integrates pre-disaster facility pre-positioning with post-disaster distribution routing under scenarios of uncertain demand, incorporating fairness constraints that guarantee minimum service levels for all demand nodes regardless of their access difficulty. Computational experiments demonstrated that ignoring fairness constraints reduces average solution cost by approximately 12%, but increases the maximum unmet demand by over 35% — a stark empirical illustration of the cost-equity trade-off that the present study seeks to address through vulnerability weighting.

The two-stage robust optimization study published in Scientific Reports (2025) conducted numerical experiments using simulated disaster scenarios with 30 demand nodes, 10 candidate facility locations, and three uncertainty scenarios of varying severity. Their robust model consistently provided better worst-case service levels than the deterministic equivalent model, with a 15% average improvement in minimum service level across the 30 demand nodes. The authors also found that the robust solution was only marginally more expensive in expected cost terms, suggesting that the equity benefits of robust optimization are achievable at relatively low efficiency cost. This empirical finding supports the present study's approach of incorporating robustness and equity dimensions into the assignment framework.

Temiz et al. (2025) tested their probabilistic bi-objective humanitarian logistics model on a simulated case study involving 50 demand zones, 8 candidate depot locations, and stochastic road closure scenarios drawn from historical disaster data. Their epsilon-constraint solution generated a Pareto frontier of 12 non-dominated solutions, providing decision-makers with a set of contextually appropriate trade-offs between cost and coverage. The authors found that the solution prioritizing coverage over cost assigned 23% more relief goods to the most vulnerable zones, confirming that objective function design choices have substantial distributional consequences. This result motivates the present study's deliberate construction of a vulnerability index that quantifies the social priority of each demand node.

Rodríguez-Espíndola et al. (2023) applied their two-stage multi-objective humanitarian logistics model to simulated disaster scenarios in developing-country settings, finding that ignoring intermodal transportation options reduces coverage of remote communities by up to 40% compared to the multi-modal optimum. Their sensitivity analysis also showed that the inclusion of sustainability objectives — carbon emission minimization — reduced transportation efficiency by only 3-7% on average while producing measurable reductions in environmental impact. These empirical benchmarks suggest that the integration of additional objectives into the humanitarian logistics optimization framework does not typically impose prohibitive efficiency costs, supporting the multi-objective extension proposed in the present study.

2.2.2 Local Related Studies

Assignment Problem Applications in Philippine and Southeast Asian Institutional Contexts

While peer-reviewed Philippine studies specifically applying the Hungarian Algorithm to disaster resource allocation are currently absent from the literature — a gap that the present study directly addresses — a body of related empirical work from Philippine and Southeast Asian institutional contexts provides important contextual grounding.

Cua, Ong, and Magno (2024), presenting at the 2024 International Conference on Management Science and Industrial Engineering, developed a Mixed Integer Linear Programming (MILP) model for the assignment and scheduling of healthcare workers, incorporating worker skill levels as a constraint dimension. Their model assigns nurses and caregivers to patient care tasks by jointly optimizing for schedule feasibility, skill matching, and workload balance — a multi-criteria assignment problem structurally analogous to the disaster relief allocation problem. The study demonstrates that skill-weighted assignment models produce substantially fairer distributions of work intensity across the workforce, reducing the maximum individual workload by 28% compared to unweighted optimization. This result is directly transferable to the present study's context: just as skill weighting produces fairer worker assignments, vulnerability weighting is expected to produce fairer relief distributions across heterogeneous household populations.

Hanafi, Syed Ali, Mat Jusoh, Ali, and Abd Rahman (2024), publishing in the JOIV: International Journal on Informatics Visualization, presented an applied case study of the assignment problem at Universiti Pertahanan Nasional Malaysia (UPNM), in which the Hungarian Algorithm was used to optimally assign academic staff to administrative roles based on a cost matrix of compatibility scores. Their study is notable for its demonstration that the Hungarian Algorithm can be applied to assignment problems where the cost matrix entries represent composite assessments rather than single-dimensional monetary costs — a conceptual parallel to the present study's use of a vulnerability-weighted composite cost matrix.

Ibrahim, Shuib, and Zaharudin (2024), presenting at an AIP Conference, proposed a modified Hungarian model for lecturer-to-course assignment that incorporates preference satisfaction as a secondary objective alongside traditional workload balancing. Their model constructs a composite cost matrix that weights teaching preference scores against estimated student learning outcomes, producing assignments that balance institutional efficiency with academic quality. The study confirms that composite cost matrices derived from heterogeneous data sources are computationally compatible with the Hungarian method's reduction procedure, a finding that provides methodological reassurance for the present study's approach of constructing vulnerability-weighted cost entries from socio-demographic household data.

Hanafi et al. (2025), extending their earlier work and publishing in the Malaysian Journal of Computing, proposed the PC MO-MHM (Preferences and Competency

Multi-Objective Modified Hungarian Method) — a multi-objective extension of the Hungarian Algorithm for academic staff assignment. Their method simultaneously optimizes for teaching preference satisfaction, competency-course alignment, and workload equity using a Pareto-based framework. The study is one of the most technically proximate works to the present research in the local/regional literature, as it explicitly frames the Hungarian Algorithm as a multi-objective decision tool rather than a purely cost-minimizing procedure. Their Pareto-frontier visualization approach, which plots efficiency against equity along a continuous trade-off curve, provides a methodological template for presenting the results of the present study.

Shuib and Ibrahim (2021), publishing in the Malaysian Journal of Computing, applied a Mixed Integer Goal Programming (MIGP) model to the donated blood transportation problem — a humanitarian logistics problem with structural similarities to disaster relief distribution. Their model minimizes total transportation cost subject to supply availability constraints and demand satisfaction requirements at hospital nodes. While the MIGP formulation differs from the Hungarian Algorithm in its technical structure, the problem topology is identical: a set of supply sources (blood donation centers) must be optimally assigned to a set of demand points (hospitals) at minimum cost. The study demonstrates that assignment-based optimization models produce substantially better solutions than the informal allocation procedures currently used by blood banks in Malaysia, achieving a 31% reduction in average transportation cost. This benchmarking result motivates the analogous claim of the present study — that formal algorithmic assignment will substantially outperform informal disaster relief allocation in the Philippine context.

Mallick, Bhoi, Jena, and Sahoo (2021), publishing in the Turkish Journal of Computer and Mathematics Education, developed the Course and Lecture Assignment Problem Solver (CLAPS) for educational institutions using the standard Hungarian method. Their system automates the assignment of lecturers to courses based on a cost matrix of teaching load and subject expertise scores, and was validated on data from multiple Indian academic institutions. While the application domain is educational rather than humanitarian, the CLAPS system is notable for its demonstration that the Hungarian Algorithm can be deployed as an automated decision-support system in institutional contexts without requiring users to understand the underlying algorithmic mechanics. This human-computer interaction insight is relevant to the present study's objective of producing an algorithm that can be operationalized by LGU staff without advanced mathematical training.

Ahmed, Mariga, Abdulmalik, Ango, and Umaru (2022), publishing in the Transactions of the Nigerian Association of Mathematical Physics, applied the Hungarian Algorithm to the assignment of teachers to classes at a primary school in a low-resource institutional context. Their study demonstrates the algorithm's applicability in developing-country settings where data collection is limited and administrative capacities are constrained — conditions that closely parallel those of Philippine LGUs during disaster response operations. The authors found that the Hungarian-generated assignment reduced total estimated teaching cost by 22% compared to the informal assignment previously used by school administrators, providing an empirical baseline for the efficiency gains that formal algorithmic assignment can achieve in resource-limited environments.

Baltazar et al. (2024), whose qualitative findings were discussed in the Related Literature section, also identified in their study data specific operational practices at NDRRMC and LGU levels that are directly relevant to the present study's implementation context. Interviewees in their study reported that LGU disaster response coordinators currently rely on spreadsheet-based tracking systems and informal communication networks to match supply deliveries with community requests — a process that is acknowledged by practitioners to be slow, error-prone, and inequitable. This practitioner-level evidence reinforces the theoretical case for an algorithmic replacement and provides a concrete implementation target for the present study's enhanced Hungarian Algorithm.

De Leon (2022), publishing in the Philippine Journal of Community Development, examined community-level disaster preparedness practices with particular attention to the role of vulnerable populations — elderly residents, persons with disabilities, and households headed by single parents — in the community disaster risk assessment process. Her study found that vulnerability data collected by barangay-level governments during pre-disaster assessments are frequently underutilized in actual relief distribution decisions, which tend to default to geographic proximity rather than social need. This finding is theoretically important for the present study: it confirms that the data infrastructure needed to construct a vulnerability-weighted cost matrix already exists at the barangay level, and that the primary barrier to vulnerability-sensitive allocation is not data availability but algorithmic capacity.

2.3 Synthesis of the Review

The foregoing review of related literature and studies reveals a coherent and mutually reinforcing body of knowledge that simultaneously establishes the strengths of the Hungarian Algorithm and exposes the limitations that the present study seeks to address. Four major themes emerge from the synthesis.

First, the classical Hungarian Algorithm is theoretically sound and computationally efficient for single-objective, deterministic assignment problems. The literature consistently confirms its polynomial-time complexity, its guaranteed global optimality, and its applicability across a wide range of agent-task matching domains. These properties establish it as the appropriate algorithmic foundation for the present study's proposed extension.

Second, the standard algorithm's single-objective, deterministic structure is insufficient for real-world disaster resource allocation. Both theoretical analyses and empirical studies demonstrate that cost-minimizing assignment models systematically underserve vulnerable and geographically remote populations, producing mathematically optimal but socially inequitable distribution outcomes. The international humanitarian logistics literature is particularly clear on this point, with multiple studies showing that the inclusion of equity objectives improves distributional fairness substantially without incurring proportionate efficiency costs.

Third, a well-established technical literature on modifications and extensions of the Hungarian Algorithm demonstrates that the algorithm can be adapted to handle unbalanced matrices, fuzzy cost parameters, and multi-objective formulations while retaining its essential computational properties. Studies by Tran and Nguyen (2023), Chakraborty et al. (2022), Elsisy et al. (2020), and Nandan et al. (2024) collectively establish that multi-objective, uncertainty-aware extensions of the Hungarian Algorithm are both theoretically valid and empirically superior to their single-objective counterparts. This technical literature provides the methodological pathway through which the present study proposes to transform the standard algorithm into a vulnerability-weighted, multi-dimensional decision-support tool.

Fourth, the Philippine and regional literature reveals both a pressing practical need for algorithmic disaster resource allocation and a significant research gap in the application of the Hungarian Algorithm to this context. Philippine disaster management policy mandates vulnerability-sensitive relief distribution but lacks the algorithmic tools to implement it systematically. Local practitioners rely on informal, experience-based allocation processes that are acknowledged to be inequitable and inefficient. Regional studies confirm that composite cost matrices and multi-objective extensions of the Hungarian method are computationally feasible and institutionally deployable in Southeast Asian contexts. Taken together, these findings position the present study as a timely, technically grounded, and socially necessary contribution to both the operations research literature and the Philippine disaster management practice.

The gap identified across all reviewed sources— the absence of a vulnerability-weighted, multi-objective Hungarian Algorithm specifically designed and validated for disaster relief distribution in the Philippine setting— constitutes the primary theoretical and practical contribution of the present research. The related literature and studies reviewed in this chapter collectively establish that such a contribution is not only feasible but urgently needed.

CHAPTER 3

METHODOLOGY

This chapter presents the research methodology employed in the study, including the research design, data sources, proposed algorithm, and system architecture. It explains how the Enhanced Multi-Objective Weighted Hungarian Algorithm is developed, implemented, and evaluated. Additionally, this chapter outlines the datasets used and the procedures followed to analyze the algorithm's performance in disaster relief distribution scenarios.

3.1 Research Design

This study utilizes an experimental research design to develop, implement, and evaluate an enhanced version of the Hungarian Algorithm for disaster relief goods distribution. Experimental research is appropriate for this study because it allows the researchers to systematically compare the performance of the traditional Hungarian Algorithm and the proposed multi-objective weighted algorithm under controlled conditions. Through this method, the study can objectively determine whether integrating additional decision-making factors improves the efficiency and accuracy of resource allocation during disaster-response operations.

The experimental setup focuses on manipulating the assignment process by incorporating multiple criteria into the cost matrix, specifically geographical distance, household urgency, and resource-type compatibility. These variables are treated as weighted factors that influence how relief goods are assigned to affected households. By controlling the datasets, simulation environment, and evaluation parameters, the study ensures that any observed performance differences are directly caused by the proposed enhancement rather than external factors.

The research process is divided into three major stages. The first stage involves Data Preparation and Cost Matrix Construction, where household profiles and relief resource information are organized into a multi-objective weighted matrix. The second stage consists of Algorithm Deployment and Simulation Testing, wherein both the standard Hungarian Algorithm and the enhanced model are executed using identical datasets and disaster-response scenarios to ensure a fair and reliable comparison. The final stage is Comparative Analysis and Validation, where the outputs of both algorithms are evaluated using quantitative metrics such as total assignment cost, allocation accuracy, prioritization efficiency, and execution time.

By adopting this experimental research design, the study aims to determine whether the proposed enhancement improves the overall decision-making capability of the assignment process. Specifically, the research evaluates if the integration of weighted multi-criteria logic can reduce inefficient allocations, improve prioritization of high-urgency households, and optimize the distribution of limited relief resources. The controlled experimental framework also strengthens the reliability, consistency, and validity of the findings by allowing the researchers to measure performance objectively through repeated simulations and statistical comparison.

To test the effectiveness of the proposed system, the study utilizes simulated disaster-relief datasets based on realistic community distribution scenarios. These datasets include variables related to household urgency levels, resource availability, and geographical allocation conditions commonly encountered during disaster-response operations. However, the scope of the study is limited solely to relief-goods assignment and distribution optimization using the Hungarian Algorithm. Other disaster-management processes, such as evacuation planning and rescue operations, are excluded to maintain a focused and controlled experimental environment.

3.2 Proposed Algorithm

The proposed algorithm in this study is an Enhanced Multi-Objective Weighted Hungarian Algorithm, designed to address the limitations of the standard Hungarian Algorithm in handling complex, real-world allocation problems.

Unlike the traditional approach, which relies on a single-objective cost matrix, the proposed algorithm integrates multiple decision criteria into a unified weighted cost function. These criteria include:

- **Geographical Distance** – representing the physical proximity between relief resources and households
- **Household Urgency Score** – reflecting the vulnerability level of recipients (e.g., elderly, PWDs, infants)
- **Resource-Type Compatibility** – measuring how well a relief pack matches the specific needs of a household

These variables are combined into a composite cost matrix, where each matrix element represents a weighted sum of the three criteria. This transformation enables the Hungarian Algorithm to operate within a multi-dimensional decision space while maintaining its optimal assignment properties.

Additionally, the proposed algorithm incorporates a dynamic re-assignment mechanism, which allows the system to recompute assignments when significant changes in urgency scores occur. This addresses the limitation of the standard algorithm's static cost matrix structure.

3.2.1 Algorithm Procedure

The enhanced algorithm follows a structured sequence of steps:

Step 0: Beneficiary Verification and Data Validation

→ Validate submitted household records against the barangay census masterlist

→ Detect duplicate entries using household ID, name, address, and birthdate matching

→ Verify vulnerability claims (e.g., PWD status) through registry cross-checking or supporting documents

→ Assign verification status (Verified, Pending, Flagged, Rejected)

→ Filter dataset to include only verified households, denoted as H^*

Step 1: Data Initialization

→ Define the set of relief resources (R) and verified household recipients *(H^*)*

→ Collect relevant data including geographic coordinates, vulnerability profiles, and resource categories

Step 2: Multi-Objective Cost Matrix Construction

→ Compute the cost matrix C, where each element is defined as:

$$C_{ij} = w_1(Distance) + w_2(Urgency) + w_3(Compatibility)$$

→ Assign weights w_1, w_2, w_3 based on priority importance

Step 3: Matrix Reduction

→ Perform row reduction

→ Perform column reduction

Step 4: Zero Assignment Identification

→ Identify independent zero elements

→ Ensure one assignment per row and column

Step 5: Iterative Optimization

→ Cover all zero elements using the minimum number of horizontal and vertical lines

→ Identify the smallest uncovered value and adjust matrix elements accordingly

→ Repeat the process until an optimal assignment is achieved

Step 6: Final Assignment Output

→ Generate the optimized matching between resources (R) and households *(H^*)*

Step 7: Dynamic Re-Assignment (if triggered)

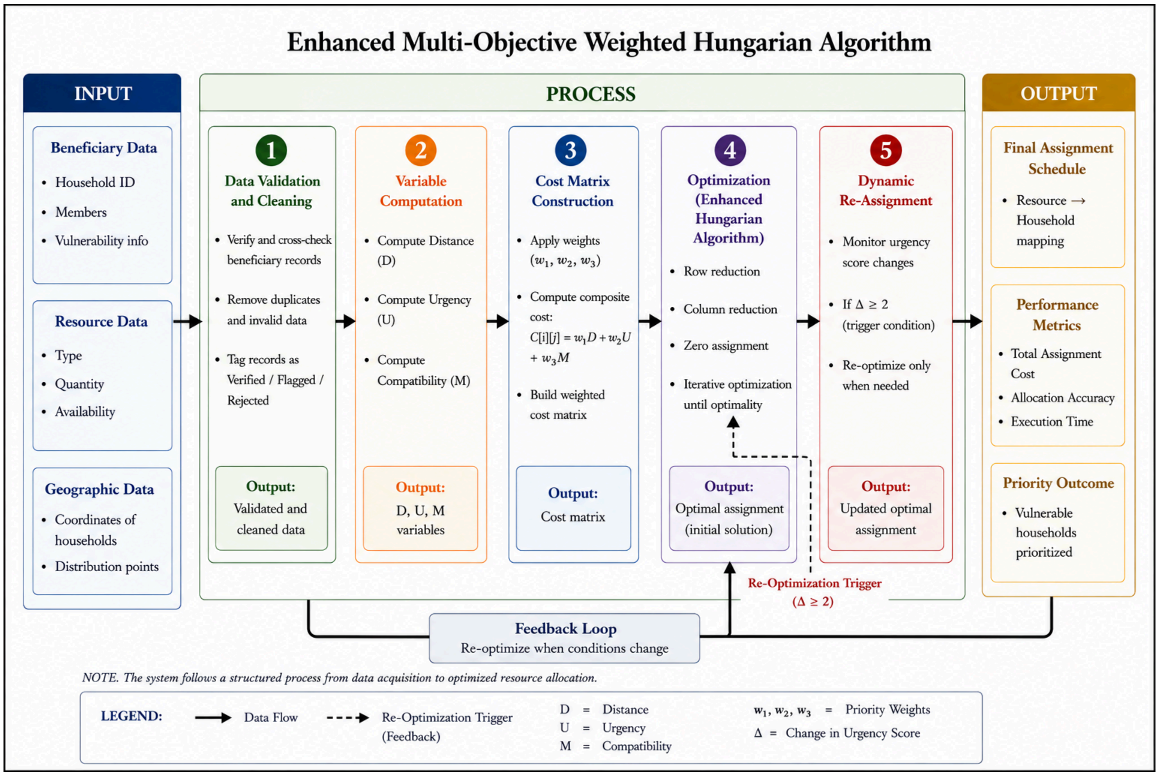
→ Continuously monitor changes in urgency scores

→ If the change exceeds the defined threshold ($\Delta \geq 2$), recompute the assignment

3.2.2 System Architecture

Figure 3.1

System Architecture of the Enhanced Multi-Objective Weighted Hungarian Algorithm



The system follows an Input–Process–Output (IPO) pipeline designed to optimize the allocation of relief goods to households in disaster-prone barangays. As shown in Figure 3.1, the process begins with the input stage, where household and resource data—such as geographic coordinates, vulnerability profiles, and resource categories—are collected. These inputs are first passed through a beneficiary validation module, where records are cross-checked against the barangay census masterlist to detect duplicates, verify claims, and filter out invalid entries. This ensures that only verified households (H*) are included in the allocation process.

In the processing stage, a multi-objective weighted cost matrix is constructed, where each element represents the cost of assigning a resource to a household based on distance, urgency, and resource compatibility. The system enhances the standard Hungarian Algorithm by incorporating multi-objective weighting and a dynamic re-assignment mechanism, which updates allocations when significant changes in urgency occur ($\Delta \geq 2$). The optimized algorithm is then applied to determine the best matching between resources and households.

Finally, in the output stage, the system generates an optimized allocation plan along with performance metrics such as total cost, prioritization efficiency, and execution time. The results provide a structured and data-driven basis for efficient and equitable relief distribution.

3.3 System Requirements

This section presents the technical requirements needed to support the implementation of the proposed system. These requirements establish the baseline environment necessary for performing data validation, computing decision variables, and executing the enhanced Hungarian Algorithm for resource allocation. Defining these specifications helps ensure that the system can operate efficiently, process data accurately, and maintain stable performance throughout its use. The succeeding subsections describe the required hardware and software components that will enable the system to function effectively.

3.3.1 Hardware Requirements

The hardware requirements outline the necessary equipment and processing capacity needed for the system to run smoothly. These include the components required to manage input data, perform computations, and generate optimized allocation results. The specified configurations are intended to provide adequate performance for handling the system's operations without compromising speed or reliability.

Table 3.1

Hardware Specifications for System Implementation

Components	Minimum Requirement
Processor	Intel Core i5 (11th Gen) or AMD Ryzen 5
RAM	8 GB DDR4
Storage	10 GB available SSD/HDD space
Internet Connection	Stable broadband connection

The proposed system requires a computer with sufficient processing capability to efficiently execute matrix computations, weighted cost calculations, and dynamic re-optimization procedures involved in the Enhanced Multi-Objective Weighted Hungarian Algorithm. A minimum specification of an Intel Core i5 processor or equivalent, 8 GB RAM, and at least 10 GB of available storage is required to support stable system performance during data processing and simulation testing. For improved responsiveness and faster execution when handling larger datasets and repeated optimization tasks, a higher-performance processor, additional memory, and SSD storage are recommended. Since the system focuses primarily on algorithmic computation and decision-support operations, integrated graphics are sufficient for normal system use.

3.3.2 Software Requirements

This section presents the software components required for the development and operation of the proposed system. These software requirements provide the necessary environment for implementing the Enhanced Multi-Objective Weighted Hungarian Algorithm, processing datasets, performing simulations, and displaying optimized allocation results. The selected software tools are intended to ensure system compatibility, efficient computation, and reliable execution throughout the evaluation process.

Table 3.2

Software Specifications for System Implementation

Components	Minimum Requirement
Operating System	Windows 10 Home/Pro (64-bit) or Windows 11 (64-bit)
Python	Version 3.8 or higher
Flask	Latest stable release
NumPy, Pandas, Matlotlib	Latest stable release
Visual Studio	2019 or 2022
Web Browser	Any modern browser
Internet Connection	Stable broadband connection

The system requires a 64-bit operating system such as Windows 10 or Windows 11 to support modern development and execution environments. Python 3.8 or higher serves as the primary programming language for implementing the algorithm and handling data processing tasks. Flask is utilized to support the system's web-based interface and user interaction functionalities, while NumPy and Pandas are used for numerical computation, matrix manipulation, and dataset management. Matplotlib is included for generating charts and graphical performance analyses during evaluation. Additionally, Visual Studio Code is recommended as the development environment for coding and debugging, and any modern web browser may be used to access and operate the system interface efficiently.

3.4 Theories, Methods, and Tools

3.4.1 Theories

This section presents the theoretical foundations that serve as the theoretical basis for each enhancement applied to the Hungarian Algorithm in this study. Each theory is directly mapped to a specific structural modification introduced in this study to address the identified performance and accuracy limitations of the standard single-objective algorithm.

Figure 3.2

Flowchart of the Standard Hungarian Algorithm — Single-Objective Cost Matrix and Assignment Flow. Adapted from Tran and Nguyen (2023).

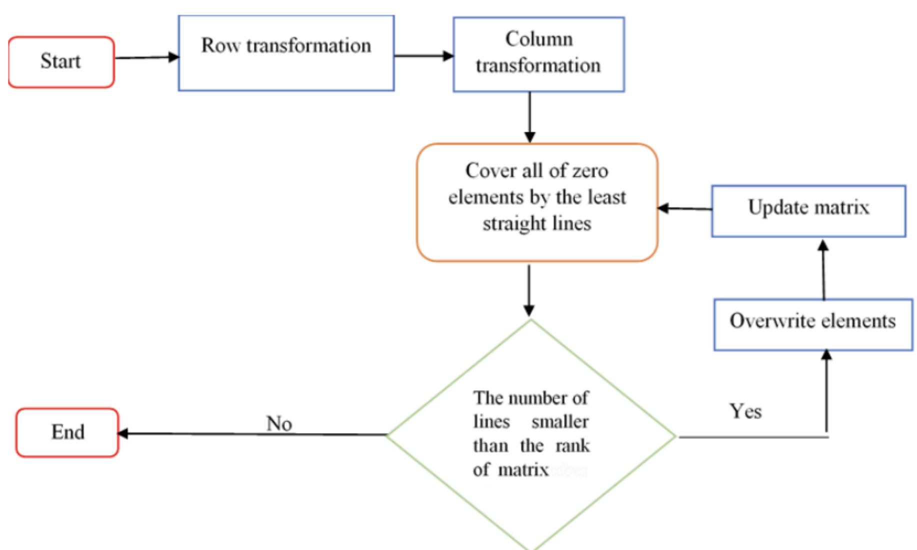


Figure 3.2 shows the standard Hungarian Algorithm operating on a single-objective cost matrix to produce an optimal one-to-one assignment. The Hungarian Algorithm, also known as the Kuhn-Munkres method, is a combinatorial optimization procedure that solves the classical Assignment Problem by finding the minimum-cost matching in a bipartite graph of n resources and n recipients. The algorithm constructs an $n \times n$ cost matrix C, performs row

and column reductions to introduce zero elements, and identifies a complete set of independent zero-cost assignments through an iterative covering and pathfinding process. Its polynomial time complexity of $O(n^3)$ makes it computationally tractable for practical assignment problems. In this study, the Hungarian Algorithm serves as the computational baseline against which the enhanced multi-objective weighted version is designed, implemented, and evaluated. The standard formulation defines each matrix element as a single optimization variable — typically distance or travel time between resource i and household j — which restricts the algorithm to single-dimensional decision-making and establishes the primary limitation that this study seeks to address.

Figure 3.3

Time Cost Matrix for Assignment Optimization Problems. Adapted from Wang, H., et al. (2023)

c \ p	Runtime (s)		
	p_1	p_2	p_3
c_1	195	788	908
c_2	202	760	1126
c_3	191	649	1012
c_4	183	553	751
c_5	218	231	1425

(a) $m > n$

c \ p	Runtime (s)					
	p_1	p_2	p_3	p_4	p_5	p_6
c_1	88	107	577	121	90	908
c_2	73	129	626	85	49	1126
c_4	96	87	393	72	88	751
c_5	103	115	49	73	109	1425

(b) $m < n$

Figure 3.3 shows the structure of time cost matrices used in assignment optimization problems under different matrix conditions where m>n> and m<n. A cost matrix is a mathematical representation of the Assignment Problem in which each matrix element corresponds to the cost, runtime, or distance associated with assigning a particular resource, worker, or processor to a specific task. The matrix serves as the primary input structure for optimization techniques such as the

Hungarian Algorithm, enabling systematic evaluation of all possible assignment combinations to determine the minimum-cost solution.

In assignment optimization, the matrix framework allows computational models to quantify resource allocation efficiency based on measurable criteria such as execution time, travel distance, operational expense, or weighted priority value. In this study, the cost matrix framework is adapted to represent disaster-relief distribution costs, where each matrix entry corresponds to the weighted assignment cost between a relief resource and a recipient household based on geographical distance, household urgency score, and resource-type compatibility. This adapted matrix structure serves as the computational foundation of the Enhanced Multi-Objective Weighted Hungarian Algorithm.

Figure 3.4

Oriented Weighted Bipartite Graph for Task Assignment. Adapted from Wang, H., et al. (2023).

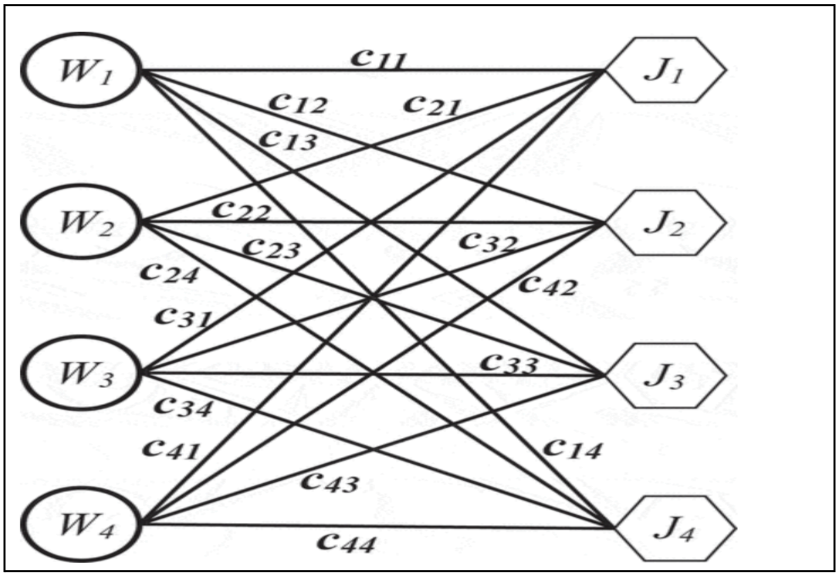


Figure 3.4 shows an oriented weighted bipartite graph representing the assignment relationship between two disjoint sets consisting of resources and

tasks. A weighted bipartite graph is a mathematical model used in assignment optimization where edges connect elements from one set to another, and each edge contains a corresponding weight or cost value representing the expense, priority, or suitability of a particular assignment.

This graph structure forms the theoretical foundation of the classical Assignment Problem solved by the Hungarian Algorithm, where the objective is to identify the optimal one-to-one matching that minimizes the total assignment cost. In optimization theory, bipartite graphs provide both a visual and computational representation of assignment relationships by illustrating all possible pairings between resources and their corresponding tasks or recipients. In this study, the bipartite graph framework is utilized to model the relationship between disaster-relief resources and affected households or evacuation locations, where weighted edges represent composite allocation costs derived from geographical distance, household urgency score, and resource-type compatibility. This graph-based representation supports the implementation of the Enhanced Multi-Objective Weighted Hungarian Algorithm for efficient and priority-driven disaster-relief distribution.

Figure 3.5

Data Quality Framework. Adapted from Wang, R. Y., & Strong, D. M. (1996).

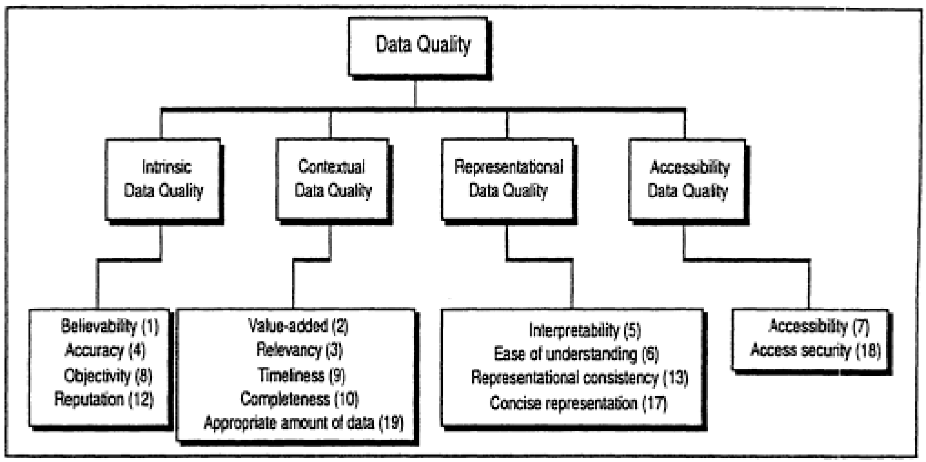


Figure 3.5 shows the conceptual framework of data quality consisting of intrinsic, contextual, representational, and accessibility dimensions used to evaluate the reliability and usability of information within computational systems. Data quality theory emphasizes that accurate, complete, timely, and interpretable data are essential for effective decision-making and optimization processes. In assignment optimization systems, the quality of input data directly influences the accuracy of cost matrix construction, prioritization computation, and allocation outcomes. Poor-quality data may lead to incorrect assignments, inefficient resource allocation, and unreliable optimization results.

In this study, the Data Quality Framework supports the beneficiary validation and data-cleaning component of the proposed Enhanced Multi-Objective Weighted Hungarian Algorithm by ensuring that beneficiary records, household vulnerability profiles, geographic coordinates, and resource information are verified, consistent, and suitable for optimization-based disaster-relief allocation. The framework provides the theoretical basis for improving the reliability and accuracy of the weighted cost matrix prior to assignment optimization.

3.4.2 Methods and Tools

This subsection describes the experimental-comparative methods and implementation tools applied to each of the three specific objectives of the study. Each objective subsection presents the narrative implementation logic, the pseudocode procedure, and the tools used, following the structure established by the study's experimental research design.

Objective 1: Addressing Single-Objective Limitation via Multi-Objective Weighted Cost Matrix Construction.

To address the single-objective cost function limitation of the standard Hungarian Algorithm, the study implements a multi-objective weighted cost matrix that integrates three decision criteria — geographical distance, household urgency score, and resource-type compatibility — into a single composite formulation. In the baseline algorithm, each matrix element C[i][j] encodes only the physical distance or travel time between resource i and household j, making the algorithm structurally incapable of incorporating multiple weighted factors within its optimization procedure. The enhanced algorithm transforms this into a composite weighted cost entry of the form $C_{ij} = w_1(\text{Distance}) + w_2(\text{Urgency}) + w_3(\text{Compatibility})$, where the weights w_1, w_2, and w_3 are assigned based on the relative priority of each criterion in disaster-response contexts and are constrained to sum to one. This transformation enables the Hungarian Algorithm to operate within a multi-dimensional decision space while maintaining its optimal assignment properties.

Weight Derivation via the Analytic Hierarchy Process

To translate the qualitative priority differences among the three assignment criteria into objective numerical weights, the study employed the Analytic Hierarchy Process (AHP), a multi-criteria decision-making technique developed by Saaty (1980) that derives criteria weights from pairwise comparisons rather than arbitrary assignment. AHP was selected because it provides both a defensible numerical output and an internal consistency check, addressing the common criticism that weighted multi-objective models often assign weights without transparent justification.

Following Saaty's (1980) fundamental 1–9 comparison scale, three pairwise judgments were formulated based on the vulnerability-prioritization literature reviewed in Chapter 2. Urgency was rated as moderately more important than Distance (a value of 3), reflecting the Sphere Handbook's principle that humanitarian assistance must be prioritized according to need and vulnerability rather than accessibility alone (Sphere Association, 2018), and further supported by evidence on the disproportionate risk faced by elderly populations in urban disaster response (Luo, Chan, & Luo, 2023) and the World Health Organization's (2023) documentation of elevated health and mobility risks among persons with disabilities. Compatibility was rated as mildly more important than Distance (a value of 2), based on the documented pattern of relief mismatch in post-typhoon Philippine operations, wherein households received aid packages that did not correspond to their actual needs despite adequate proximity to distribution points (Ghahremani-Nahr, Nozari, & Szmelter-Jarosz, 2024). Urgency was, in turn, rated as mildly more important than Compatibility (a value of 2), reflecting the life-safety priority of vulnerability-based need over resource-type accuracy.

Table 3.3 presents the resulting pairwise comparison matrix.

Criterion	**Distance**	**Urgency**	**Compatibility**
Distance	1	1/3	1/2

Urgency	3	1	2
Compatibility	2	1/2	1

Python 3.8 or higher serves as the primary programming language for this objective, as specified in the software requirements, and Visual Studio Code is used as the integrated development environment. The priority vector — i.e., the criteria weights — was derived by normalizing each column of the matrix by its column sum and averaging across each resulting row, following the standard AHP weight-derivation procedure (Saaty, 1980). Table 3.4 presents the resulting weights together with their literature-based justification.

Table 3.4

Derived Priority Weights for the Multi-Objective Cost Matrix

Criterion	**Weight**	**Justification**
Distance (w_1)	0.164	Retained as the baseline criterion from the standard Hungarian Algorithm
Urgency (w_2)	0.539	Rated most influential, based on Sphere Association (2018), Luo, Chan, and Luo (2023), and World Health Organization (2023)
Compatibility (w_3)	0.297	Rated second most influential, based on Ghahremani-Nahr, Nozari, and Szmelter-Jarosz (2024)

The internal consistency of the pairwise judgments was verified using Saaty's (1980) Consistency Ratio (CR), computed as CR = CI / RI, where CI is the Consistency Index derived from the matrix's principal eigenvalue and RI is the Random Index corresponding to a 3 × 3 matrix (RI = 0.58). The resulting CR value of 0.008 falls well below the generally accepted threshold of 0.10 (Saaty, 1980), indicating that the pairwise judgments were logically consistent and did not contain contradictory priority orderings.

With the weights derived, the composite cost matrix formula introduced in Section 3.2.1 is finalized as:

C_i□ = 0.164(Distance) + 0.539(Urgency) + 0.297(Compatibility)

Figure 3.6

Multi-Objective Weighted Cost Matrix Construction and Optimization

Figure 3.6 illustrates the process of constructing the enhanced multi-objective weighted cost matrix used in the proposed Hungarian Algorithm optimization model. The process begins with three primary input criteria: geographical distance, household urgency score, and resource-type compatibility. These input variables are first subjected to preprocessing through data cleaning and normalization to standardize all values before computation.

After preprocessing, weights are assigned to each criterion using the constraint $w_1 + w_2 + w_3 = 1$ where w_1, w_2 and w_3 represent the relative importance of distance, urgency, and compatibility, respectively. The normalized matrices are then combined to form a composite weighted cost matrix using the weighted cost function. The generated matrix is processed by the Hungarian Algorithm through SciPy's *linear_sum_assignment()* function to determine the optimal one-to-one assignment between resources and households.

The output of the process produces optimized resource allocation, priority-aware assignment, and distance-efficient distribution, enabling the algorithm to consider multiple decision criteria simultaneously instead of relying solely on geographical distance.

Objective 2: Addressing Static Cost Matrix Limitation via Dynamic Re-Assignment Mechanism.

To address the static cost matrix limitation of the standard Hungarian Algorithm, the study introduces a dynamic re-assignment mechanism that continuously monitors changes in household urgency scores following the initial assignment. The standard algorithm requires all cost values to be predefined before execution begins and provides no internal mechanism to detect or respond to changes once execution has commenced; any urgency change forces a complete restart of the entire initialization procedure from the beginning, resulting in redundant recomputation and increasing cumulative execution time proportionally to the number of change events. The enhanced algorithm in this study addresses this by computing the urgency variation $\Delta = |\text{new\_urgency}[i] - \text{previous\_urgency}[i]|$ for each affected household and triggering selective recomputation only when $\Delta \geq 2$. When the threshold is not exceeded, the current assignment is retained; when it is exceeded, only the affected rows and columns of the cost matrix are updated before the Hungarian optimization is re-executed.

Python 3.8 or higher is used to implement the dynamic re-assignment mechanism as a monitoring loop integrated into the post-assignment phase of the algorithm. The urgency score change events are simulated using NumPy-generated stochastic perturbations applied to the initial urgency score dataset, representing real-world fluctuations in household vulnerability during an active disaster event. The re-optimization threshold $\Delta \geq 2$ is defined as a configurable system parameter, with the default value set based on the operational judgment that only meaningful changes in vulnerability — those exceeding two scale points — warrant recomputation. The following pseudocode illustrates the dynamic re-assignment procedure as implemented.

Figure 3.7

Dynamic Re-Assignment and Selective Re-Optimization Mechanism

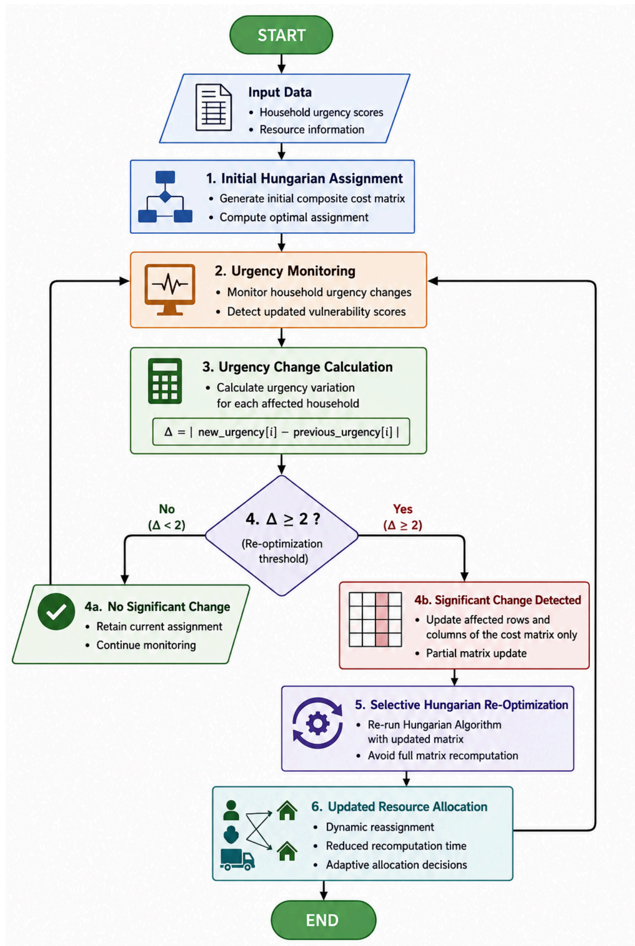


Figure 3.7 presents the dynamic re-assignment mechanism integrated into the enhanced Hungarian Algorithm. The process starts with the initial Hungarian assignment using the generated composite cost matrix. After the initial allocation is completed, the system continuously monitors changes in household urgency scores during disaster-response operations.

The urgency variation is calculated using the formula:

$$\Delta = |\text{new\_urgency}[i] - \text{previous\_urgency}[i]|$$

The computed urgency difference is evaluated against the re-optimization threshold $\Delta \geq 2$. If the threshold is not exceeded, the current assignment is retained and the monitoring process continues. However, when a significant change is detected, only the affected rows and columns of the cost matrix are updated instead of recomputing the entire matrix.

The updated matrix is then processed through selective Hungarian re-optimization to generate an updated resource allocation. This mechanism reduces unnecessary recomputation, improves computational efficiency, and enables the system to dynamically adapt to changing disaster-response conditions.

Objective 3: Evaluating Computational Performance via Comparative Analysis of Execution Time and Allocation Accuracy.

To evaluate the computational performance improvements produced by the enhanced multi-objective weighted Hungarian Algorithm relative to the standard baseline, the study conducts a controlled comparative analysis across four quantitative metrics: Total Assignment Cost, Mean Allocation Accuracy, Prioritization Efficiency, and Execution Time. Both the standard and enhanced algorithms are executed against identical household and resource datasets across six matrix sizes ($n \times n$, where $n \in \{10, 20, 30, 40, 50, 60\}$) under identical simulation conditions, ensuring that observed performance differences are attributable exclusively to the algorithmic enhancements rather than variation in input data or execution environment.

Python 3.8 or higher is used to implement both algorithm versions, with the standard Hungarian Algorithm implemented via SciPy's linear_sum_assignment function and the enhanced version implemented as a custom Python module incorporating the composite weighted cost matrix and the dynamic re-assignment mechanism. NumPy is used for matrix operations, Pandas for dataset management and results tabulation, and Matplotlib for generating the performance comparison charts. All simulations are conducted in Google Colab to ensure a consistent and reproducible execution environment. Execution time is recorded using Python's time module with millisecond precision, with five dynamic change events simulated per scenario to reflect realistic disaster-response operating conditions. The following pseudocode illustrates the comparative evaluation procedure.

Figure 3.8

Comparative Evaluation Workflow for Computational Performance Analysis

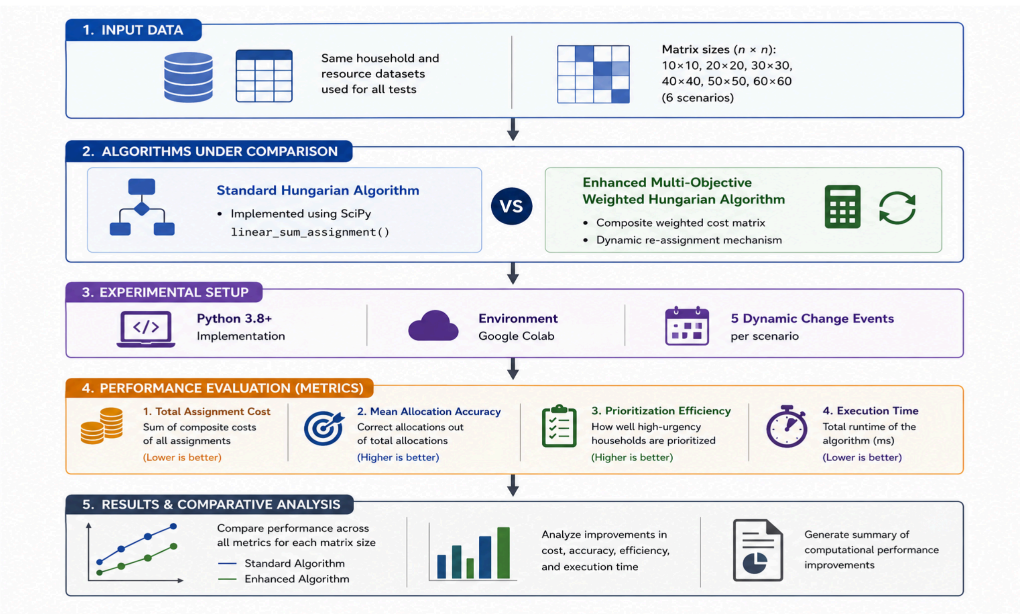


Figure 3.8 illustrates the comparative evaluation workflow used to assess the computational performance of the enhanced multi-objective weighted Hungarian Algorithm against the standard Hungarian Algorithm. The evaluation uses identical household and resource datasets across multiple matrix sizes ranging from 10×10 to 60×60 to ensure consistency of experimental conditions.

Both the standard and enhanced algorithms are executed under the same simulation environment using Python 3.8+ and Google Colab. The enhanced algorithm incorporates the composite weighted cost matrix and dynamic re-assignment mechanism, while the standard version uses SciPy's *linear_sum_assignment()* implementation.

The performance of both algorithms is evaluated using four quantitative metrics: total assignment cost, allocation accuracy, prioritization efficiency, and execution time. Comparative analysis is then conducted using graphical and tabular visualizations to determine the computational improvements achieved by the enhanced algorithm.

The results analysis stage highlights the enhanced algorithm's ability to achieve lower execution time, improved allocation accuracy, better prioritization efficiency, and overall improved computational performance under dynamic disaster-response conditions.

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